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Enhancing Route Optimization in Road Transport Systems Through Machine Learning: A Case Study of the Dakhla-Paris Corridor

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Abstract: Road transport systems (RTS) play an essential role in global supply chains, facilitating the efficient transport of goods and services over long distances and thus supporting economic activity on a worldwide scale. However, these systems face numerous challenges, particularly regarding safety, cost, and route optimization, requiring innovative and practical solutions to improve their overall performance. This paper proposes an in-depth analysis of RTS features forming a detailed dataset collected on the route between Dakhla (Morocco) and Paris (France). The study relies on applying advanced mathematical modeling techniques and analyzing several datasets to train various machine learning algorithms. The main objective is to identify optimized routes, combining high safety standards, reduced costs, and shorter transport times. The results show that the adopted approach results in safer and more efficient routes and complies with operational and regulatory constraints. Furthermore, this analysis highlights the importance of data quality and the integration of advanced technologies to deliver an intelligent route optimization system with significant reductions in cost and time. Finally, our results reveal that neural networks outperform other algorithms in this field, proving their superior effectiveness for this specific application.

Keywords: road transportation systems; route optimization; machine learning; neural networks; transport efficiency; cost reduction; optimization algorithms

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1. Introduction

Road transport is an essential part of global supply chains, facilitating the movement of goods and services over vast distances. However, this mode of transport faces significant challenges, such as congestion, safety risks, and optimization obstacles. These issues are particularly critical in dynamic environments, such as humanitarian logistics, where rapid, efficient responses are essential to saving lives. The lack of up-to-date information on traffic conditions, road quality, and safety risks exacerbates these problems, making it more difficult for logistics planners to make informed decisions [1,2]. Congested road networks lead to delays and increased fuel consumption, contributing to environmental degradation [3]. Safety risks, including accidents and offenses, threaten human life and the

safety of goods [4]. Costs, journey times, traffic density, and safety risks make it challenging to optimize transport routes [5].

Advances in artificial intelligence (AI) and machine learning hold great promise for meeting these challenges. These technologies make it possible to integrate real-time data from various sources, such as GPS, traffic sensors, and weather forecasts, to optimize transport routes dynamically [6].

1.1. Challenges in Optimizing Transport Routes

Optimizing transportation systems on a global scale is a pressing issue affecting various sectors, including trade, humanitarian, and logistics [6]. As the world becomes increasingly interconnected, the demand for efficient transportation is growing, posing operational and strategic challenges [6]. Transport optimization must consider factors such as route planning, cost efficiency, fuel consumption, and the unpredictability of events such as natural disasters or political instability [6]. In humanitarian missions, such delays can have serious consequences, affecting the safety of the personnel involved [6,7]. A multi-criteria approach incorporating advanced technologies such as AI and machine learning is needed to optimize routes and manage real-time data [6]. Transport infrastructure varies considerably worldwide, particularly in developing countries where poor road conditions and a lack of technological infrastructure hamper efficient transport operations [6]. In humanitarian aid missions, for example, access to reliable, up-to-date information can mean the difference between a successful response and a prolonged delay [6].

In addition, rising global fuel prices and the need for more sustainable transport solutions mean that optimization models must also take account of environmental impacts, particularly in long-distance logistics [6,8].

1.2. Artificial Intelligence in Route Optimization

AI and machine learning have emerged as powerful tools for addressing transport route optimization challenges [9]. These technologies enable the analysis of large datasets and the identification of patterns that would be difficult to detect manually [10]. By leveraging machine learning algorithms for real-time routing, it becomes possible to predict traffic conditions and dynamically adjust routes to enhance efficiency [11]. This capability optimizes travel times and reduces fuel consumption and emissions, contributing to more sustainable transportation systems[12].

Recent studies have demonstrated that AI-based approaches can significantly reduce transport costs while improving safety by simultaneously considering multiple criteria [13,14]. These criteria include traffic congestion, road conditions, weather patterns, and vehicle performance, allowing for a comprehensive optimization strategy. As a result, integrating AI and machine learning into transport systems promises to transform the way routes are planned and managed, leading to more efficient, cost-effective, and safer transportation solutions.

1.3. Objective of Our Study

This study aims to develop an optimized model for road transport between Dakhla (Morocco) and Paris (France), integrating various criteria, such as safety, cost, and travel time. The study relies on applying advanced mathematical modeling techniques and analyzing several data sources to train machine learning algorithms. Drawing on an exhaustive dataset, the study seeks to identify the most efficient machine learning algorithm for route optimization. Preliminary results indicate that this model can significantly reduce transport costs and journey times while maintaining high safety standards.

2. Related Works

As shown in Table 1, several studies have explored different methods for improving road transport efficiency and reducing risk. Multi-criteria decision-making models have been widely used to evaluate and prioritize journey time, cost, and safety factors. However, these approaches have often addressed specific objectives, such as reducing delays or CO_2 emissions, without consistently integrating safety, costs, and other constraints.

Table 1. Related Works.

Article	Objective	Result
[1]	Exploring AI applications in transportation to improve sustainability and efficiency.	Identification of various transport applications to improve sustainability and efficiency
[2]	Presenting AI-based real-time route optimization solutions.	Improving route efficiency and reducing costs with AI.
[3]	Analyzing trends and challenges in urban mobility, including the impact of traffic and congestion.	Detailed data on the impacts of urban congestion on mobility and recommendations for mitigation.
[6]	Presenting dynamic route optimization strategies based on data analysis.	Development of efficient route optimization strategies based on real-time data.
[7]	Using ML for improving traffic management through predictive traffic management.	High traffic condition prediction and route adjustment.
[11]	Examining the use of machine learning for real-time traffic prediction and route planning.	Accurate traffic prediction and improved real-time route planning.
[13]	A comprehensive review of the role of AI in enhancing transport safety and efficiency.	AI significantly improves transportation safety and efficiency.
[15]	Developing dynamic traffic assignment and simulation methodologies for advanced transportation systems.	Optimizing advanced transportation systems with practical applications.

2.1. AI Applications in Transportation

Studies [1,16] explore the broad applications of AI in transportation, focusing on sustainability and efficiency enhancements. These studies underscore the diverse implementation of AI technologies across different transportation scenarios, including traffic flow management and predictive maintenance systems. They highlight a significant improvement in operational sustainability due to optimized route planning and vehicle utilization.

2.2. Real-Time Route Optimization

Studies [2,17] present AI-driven solutions for real-time route optimization, detailing the algorithms' capability to dynamically adjust to changing traffic conditions. The results from these studies illustrate a notable reduction in journey times and operational costs, affirming the practical benefits of integrating AI in everyday routing applications.

2.3. Urban Mobility Trends

Studies [3,18] analyze the current trends and challenges in urban mobility, focusing on congestion and its impacts on city infrastructure. The comprehensive data show how

congestion affects urban mobility and the studies provide valuable recommendations for mitigating these issues through more intelligent traffic management systems.

2.4. Dynamic Route Optimization Strategies

Studies [6,7,11] collectively present a series of strategies for dynamic route optimization, employing a variety of machine learning techniques to adapt to real-time data inputs. These studies contribute to understanding how different data sources, such as real-time traffic feeds and weather information, can be integrated to enhance route decision-making processes.

2.5. Comprehensive AI Integration in Transport

Study [13] thoroughly reviews the role of AI in enhancing transport safety and efficiency, advocating a holistic approach to incorporating AI in transportation systems. This review confirms the effectiveness of AI in improving transport outcomes and sets a precedent for future research exploring deeper integration of AI.

2.6. Multi-Objective Optimization Using Evolutionary Algorithms

Though there are many works focusing on traditional route optimization techniques (Dijkstra, A*, neural networks, etc.), there are few works addressing multi-objective optimization in-depth in the case of intelligent transportation systems. This point is crucial in order to balance and reconcile typically opposing criteria, e.g., cost, time, safety, energy, and consumption. For example, ref. [19] proposed a method for constructing multi-layer road networks using a genetic algorithm with the assistance of multi-agent clustering, enabling dynamic tuning of high-capacity networks. Ref. [20] designed a novel genetic algorithm model for solving complex vehicle routing problems with perishable products, taking into account time window and quality constraints. These approaches reflect the growing attractiveness of evolutionary methods, and in particular, genetic algorithms, to achieve strong, flexible, and multi-criteria optimization in intelligent road environments.

2.7. Traffic Prediction and Management

Study [15] focuses on developing dynamic traffic assignments and simulation methodologies, crucial for advanced transportation system planning. The methods discussed are instrumental in simulating potential traffic scenarios and adjusting routes to anticipate regular and irregular traffic patterns. Each of these studies contributes to a layered understanding of how artificial intelligence and machine learning are revolutionizing transportation. By examining these works, we align our research objectives to address the gaps identified in the literature, particularly in applying advanced machine learning algorithms for route optimization between widely separated geographical locations, such as Dakhla and Paris.

This study aims to build on these foundational works by introducing an innovative approach that incorporates real-time data for route optimization and utilizes a comprehensive set of machine learning techniques to ensure the safety, efficiency, and cost-effectiveness of road transport systems.

3. Proposed Approach

Optimizing transport routes has become a significant concern in a context where efficiency, safety, and profitability are essential. As logistics requirements increase, particularly between destinations as far apart as Dakhla and Paris, advanced techniques such as machine learning are necessary to meet these challenges. As presented in Figure 1, our approach proposes an innovative model that leverages the capabilities of machine learning algorithms to analyze a variety of data in real-time to optimize routes by integrating safety,

cost, and logistical criteria. This framework meets current operational requirements and improves strategic decision-making in road transport.

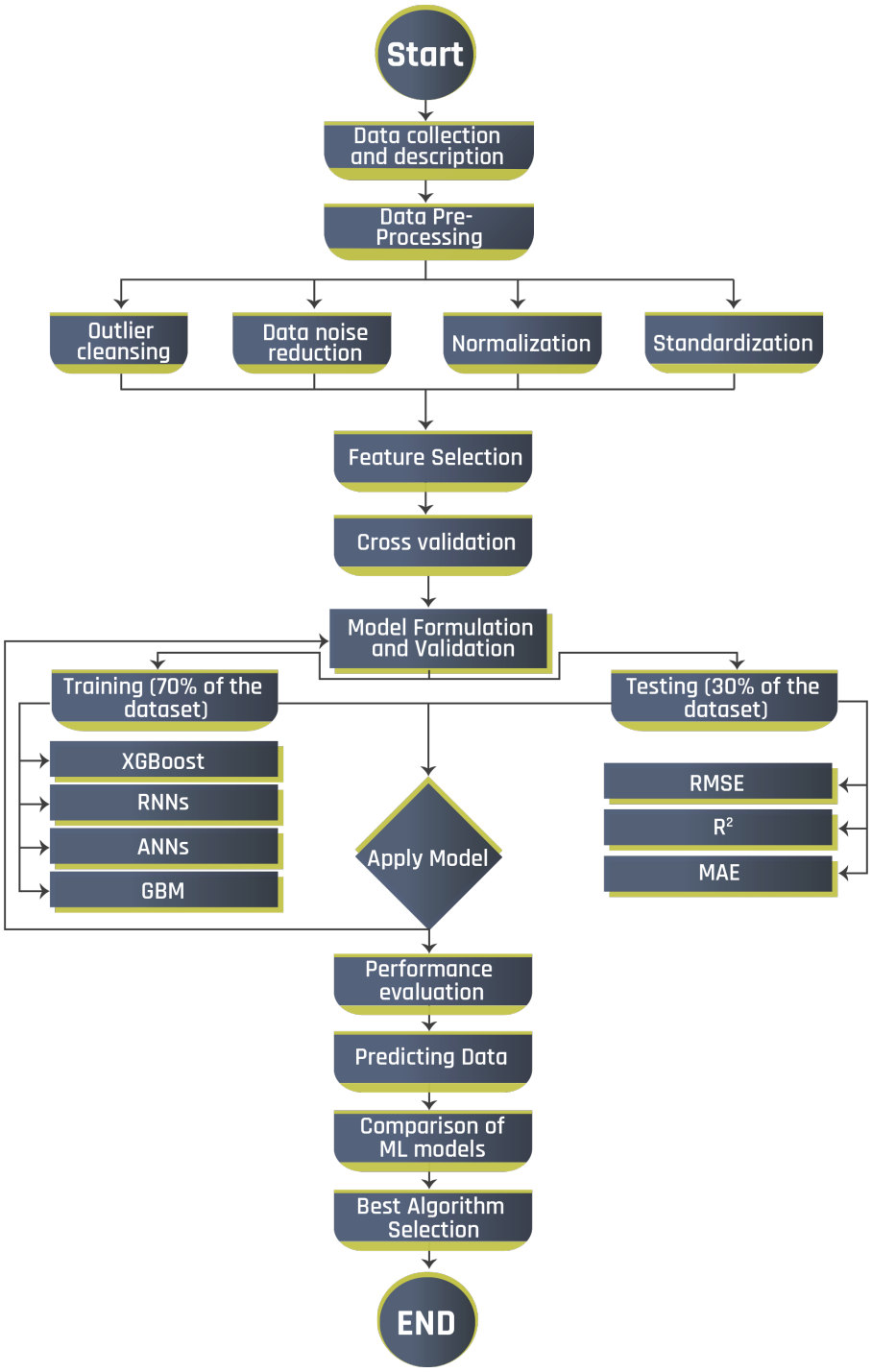


Figure 1. Methodology for the proposed approach.

3.1. Data Preprocessing

Data preprocessing consists of preparing the data so that it can be grouped in a file that can be used by machine learning. It comprises the following phases:

- Data Collection: First of all, the data must be assembled on possible paths with constraints in a consolidated form.

- Data Wrangling: Making the data usable by machine learning algorithms by extracting features through data cleaning, breakdown of data, and data scaling, as well as data shaping and transformation.

The resulting features are grouped and classified as follows:

- Duration (h): Total journey time, obtained by dividing the distance by the average speed expected on each type of road (freeway, trunk road, etc.).
- Mileage (km): The distance between the departure and destination points. Route calculation services such as Google Maps or OpenStreetMap use this information for each route.
- % Accidents: Average percentage of accidents on the route, obtained by analyzing local road safety statistics for each road segment crossed.
- % Infractions: Rate of driving infractions on the route. It is obtained from police reports and traffic control statistics, especially on high-risk roads.
- Road Type: Road category (freeway, trunk road, etc.), which is determined by the type of each route segment (A6, N4, etc.).
- Weather: Average weather conditions during the trip are based on seasonal climate databases for the time of year of the journey.
- Cost (EUR): Total estimated trip cost (fuel, tolls, etc.). The sum of fuel and toll costs.
- Fuel (L): Volume of fuel required for the trip.
- Toll (EUR): The total cost of tolls for the route. Toll cost calculators, such as those offered by ViaMichelin, can provide this information.
- Interest: Route category (logistics, emergency, etc.). Choice according to the need for the journey (goods transport, urgent personal journeys, etc.).
- Network Coverage (%): The quality of network coverage along the route. This is an estimate made from coverage maps provided by operators.
- Intelligent Route: The presence of intelligent road technologies (adaptive signaling, etc.). Local or regional road infrastructures provide this information.
- Slope (%): Average inclination of the road, calculated by analyzing the altitudes at the start and end points or by consulting topographical maps.
- Traffic Density: The expected level of traffic. It is obtained via real-time traffic applications (Waze, Google Traffic) or average density data.
- Charging Points: Number of electric charging stations along the route.
- Service Stations: Number of service stations along the route.
- Landscape: The dominant landscape (mountain, desert, coastal).
- Road Durability: Local infrastructure management data determine road durability (frequency of repairs, pavement quality).
- Road Safety: A general assessment of the route's safety (presence of speed cameras and risk zones) is obtained via road safety reports.
- Toll Waiting Time (min): The average waiting time at toll booths, estimated from local traffic statistics.
- Average Temperature (°C): The average temperature along the route, obtained from seasonal climate data for the regions crossed.
- Truck Accessibility: Indicates whether the route is accessible to trucks. Road restrictions are available from local authorities for this.
- Tunnels/Bridges: Number of tunnels and bridges on the route.
- Escape Options: There will be alternative routes in the event of closure.
- CO_2 emissions (kg): Estimated CO_2 emissions.
- Speed Limits (km/h): The maximum speed allowed on each segment by local legislation for each type of road.
- Driving Comfort: Level of comfort for the driver (road conditions, curves, etc.).

- Peak Hour: Potential peak hours along the route.
- User Choice %: This indicates the route's popularity among users. It is obtained by choice analysis or a user survey.
- Optimized: Indicator of the optimization of a transport route, equal to 0 or 1. This indicator is often based on the mathematically calculated score. If the score exceeds 0.7, then the route is optimized.

3.2. Mathematical Modeling

Our dataset is built from sensors and open-source information. Our dataset team has made a colossal effort to gather and process the data. Our work is based on calculating scores, for which we have developed a mathematical equation enabling a very sophisticated level of transport route optimization. To normalize each feature using the low-high method, we apply the following formula to each value of a feature in the dataset:

$$\text{norm_value} = \frac{v - \text{low}_v}{\text{High}_v - \text{low}_v} \tag{1}$$

where:

- norm_value: The normalized value.
- v: The original feature value.
- low_v: The minimum value of the feature.
- High_v: The maximum value of the feature.

Next, each normalized value is multiplied by the weight associated with this feature.

$$w_{\text{value},i} = \text{norm_value}_i \cdot w_i \tag{2}$$

With $\boldsymbol{w}$ being the weight coefficient associated with this feature. Finally, the sum of these products is deduced as follows:

$$\text{Sum } = \sum_{i=0}^{n} w_\text{value } i \tag{3}$$

To process the dataset and calculate the route optimization score, we need to determine the primary and secondary features, normalize them, define the weights, and then calculate the optimization score. Here is the method followed to perform this analysis:

3.2.1. Identification of Features

The main features are Duration, Mileage, % Accidents, % Infractions, Cost, Fuel, Toll, and $mathrmCO_2$ Emissions. Secondary features are Road Type, Weather, Interest, Network Coverage, Smart Route, Slope, Traffic Density, Charging Points, Service Stations, Landscape, Road Durability, Road Safety, Toll Waiting Time, Average Temperature, Truck Accessibility, Tunnels/Bridges, Escape Options, Speed Limits, Driving Comfort, and Rush Hour.

3.2.2. Value Normalization

Values are scaled, for example, using low-high normalization for quantitative features.

3.2.3. Definition of Weights

Each feature is assigned a weight according to its importance in route optimization (Duration: 30%, Cost: 25%, etc.).

3.2.4. Calculating the Optimization Score

A score indicator for each transport route is calculated using Algorithm 1, which serves as a practical implementation guide based on the previously presented formulas.

Algorithm 1: Calculating the score for each route

```
begin
    Initialize: routes as a list
    Set low_values as a dictionary;
    Set high_values as a dictionary;
    Set w as a dictionary;
    Function sc(rout);
    begin
        Set score to 0;
        foreach key in w do
            Set norm_value to (rout[key] −
             low_values[key])/(high_values[key] − low_values[key]);
            Set w_value to norm_value · w[key];
            Add w_value to score;
        return score;
    End Function;
    foreach route in routes do
        Set score to sc(rout);
        Print "Score -", rout["Name"], ":", score;
```

The implementation was carried out using Python 3.12.3. The script is programmed for this purpose. Machines performed the calculation. The final score is compared with the threshold of 0.7. Once all the steps have been completed, we have obtained the optimization score for each route. Our work on this dataset to calculate a route optimization score was a very long and demanding process in computing time and resources. While effective, the classical methods used for this optimization could not efficiently handle the large amounts of data and complex calculations required to obtain real-time results. The machine learning approach seems to be an ideal solution to overcome these limitations. By using machine learning models, it is possible to automate the optimization process while reducing processing time and improving prediction accuracy. Machine learning would enable better management of massive data and adapt routes according to actual conditions and traffic trends without requiring intensive calculations for each request.

So, replacing traditional computing with machine learning models would improve the process's efficiency and speed and offer more accurate predictions tailored to dynamic changes in road conditions. Our dataset is used for algorithm training.

3.3. Algorithms Selection

In our quest to develop an optimized model for road transport, we carefully selected machine learning algorithms based on their demonstrated performance in handling complex optimization problems and their ability to process large datasets efficiently. The algorithms chosen are pivotal for achieving high accuracy in predicting and optimizing transport routes. Here is a detailed breakdown of each algorithm and its justification:

- **Support Vector Machines (SVM)**: **Rationale:** SVMs are renowned for their robustness in classification challenges, especially in high-dimensional spaces. Their capability to model non-linear decision boundaries using kernel tricks makes them invaluable for route optimization, where the relationship between input features (like traffic and road conditions) and outcomes (optimal routes) may be highly non-linear.

Strengths: Effective in high-dimensional spaces, SVMs are less prone to overfitting, particularly in cases where the number of dimensions exceeds the number of samples.
Limitations: They require a good choice of kernel and regularization parameters, which can be computationally intensive to tune. Additionally, SVMs do not directly provide probability estimates, which are crucial for some decision-making processes.

- **Naive Bayes: Rationale:** Naive Bayes' simplicity and speed make it an excellent choice for real-time route optimization. Given its strong feature independence assumption, it performs well on large datasets with many features describing each route's conditions.
  Strengths: It is speedy for training and prediction, making it suitable for real-time analysis applications.
  Limitations: Its assumption of independent predictors can be a drawback in scenarios where the predictors are interdependent, such as traffic conditions that affect road safety.
- **Neural Networks: Rationale:** Neural networks are at the forefront of handling complex patterns and interactions between variables. They are capable of learning almost any non-linear function, which makes them particularly adept at adapting to dynamic environments like those encountered in route optimization.
  Strengths: They are highly flexible and can be trained to understand complex relationships and interactions between input features.
  Limitations: Neural networks require substantial data to train effectively and can be opaque regarding interpretability (often called "black boxes").
- **Logistic Regression: Rationale:** Despite its simplicity, logistic regression can be a robust baseline for binary classification problems. In the context of route optimization, it can efficiently handle binary decisions, such as route selection between two alternatives.
  Strengths: It is transparent, easy to implement, and provides the benefit of understanding the influence of each feature.
  Limitations: It is constrained to linear decision boundaries, which can limit its effectiveness in more complex scenarios.
- **Gradient Boosting: Rationale:** Gradient boosting is a potent ensemble technique that builds models sequentially to correct the predecessors' errors, thereby improving accuracy. Its applicability in non-linear and complex datasets where there are multifaceted relationships between variables make it an excellent choice for optimizing routes.
  Strengths: It often provides predictive accuracy that cannot be trumped by other algorithms and handles a mix of continuous and categorical data well.
  Limitations: If not tuned properly, it can be prone to overfitting and may require careful setting of parameters and stopping criteria to prevent computation inefficiencies.
  Each algorithm was selected for its technical merits and its proven applicability in transportation and logistics scenarios, ensuring that our approach is grounded in theoretical and practical considerations. By leveraging the unique strengths of these diverse algorithms, we aim to develop a robust predictive model that can dynamically adapt to varying conditions and optimize routes with high precision.

3.4. Performance Evaluation

To compare algorithm results, we used the following indicators:

- Test and score, using metrics such as precision, recall, F1-score, AUC, etc.
- A box plot is used to compare the distribution of performance scores between different algorithms.
- Matrix confusion through correct and incorrect predictions.

4. Results and Discussion

In this section, we present and analyze the results obtained from the experiments conducted. We will examine the main performances of the different models tested. Finally, a detailed comparison will be made to evaluate our approach against the few existing works.

4.1. Results Analysis

As shown in Table 2 table above shows, the gradient boosting model exhibits a moderate area under the curve (AUC) score of 0.642, indicating a limited ability to distinguish between classes. The classification accuracy (CA) is relatively high at 0.867, suggesting good overall performance. However, while decent, the F1-score of 0.836 indicates a balance between precision and recall but with room for improvement. Both precision and recall are equal at 0.867, showing that the model consistently identifies positive instances.

Table 2. Results.

Model	AUC	$\widehat{CA}$	F1	Prec	Recall
Gradient Boosting	0.642	0.867	0.836	0.807	0.867
Logistic Regression	0.840	0.867	0.881	0.904	0.867
SVM	0.728	0.900	0.853	0.810	0.900
Naive Bayes	0.630	0.933	0.918	0.938	0.933
Neural Network	0.741	0.933	0.918	0.938	0.933

Logistic regression demonstrates a strong AUC of 0.840, reflecting a high capability to distinguish between classes. The CA matches that of gradient boosting at 0.867, indicating consistent performance. Notably, the F1-score is higher at 0.881, suggesting a better balance between precision and recall. The precision is exceptionally high at 0.904, indicating that the model is very effective at identifying positive instances, though recall remains at 0.867.

The SVM model has a respectable AUC of 0.728, indicating an excellent ability to distinguish between classes. It boasts the highest CA at 0.900, showing robust overall performance. The F1-score of 0.853 reflects a balanced performance but not as high as logistic regression. Precision is somewhat lower at 0.810, suggesting some room for improvement in correctly identifying positive instances, while recall is strong at 0.900, indicating excellent detection of actual positives.

Naive Bayes shows the lowest AUC at 0.630, indicating a weaker ability to distinguish between classes than other models. However, it achieves the highest CA at 0.933, alongside neural network, indicating excellent overall performance. The F1-score of 0.918 is very high, demonstrating a good balance between precision and recall. Precision at 0.938 and recall at 0.933 suggest that the model is highly effective in identifying and detecting positive instances.

The neural network model has a solid AUC of 0.741, showcasing an excellent ability to differentiate between classes. It ties with naive Bayes for the highest CA at 0.933, indicating superior overall performance. The F1-score of 0.918, identical to that of naive Bayes, reflects an excellent balance between precision and recall. Both precision and recall are high at 0.938 and 0.933, respectively, making the neural network a robust choice for accurate classification and positive instance detection.

The analysis of these figures clearly shows that the the neural network and naive Bayes models excel in performance. Both models achieve an AUC of 0.933, an F1-score of 0.918, a precision of 0.938, and a recall of 0.933. These high scores indicate their ability to discriminate between classes and effectively minimize errors. In comparison, while

the logistic regression model also performs well, it has a lower AUC of 0.840, suggesting slightly less robustness in class separation.

Therefore, the neural network and naive Bayes models emerge as the most robust choices for optimizing accuracy and class detection in the dataset. Their high precision and recall values underline their reliability in correctly identifying positive instances and minimizing false negatives. These attributes make them highly suitable for tasks where precise classification is essential, confirming their effectiveness in practical machine-learning applications.

As shown in Figures 2 and 3, the distribution of performance scores between the neural network and naive Bayes models reveals that the neural network covers a range of the best-performing thresholds. This more precise distribution is attributed to its higher AUC, suggesting a more robust ability to distinguish between classes. Consequently, the neural network demonstrates a slightly superior classification capability.

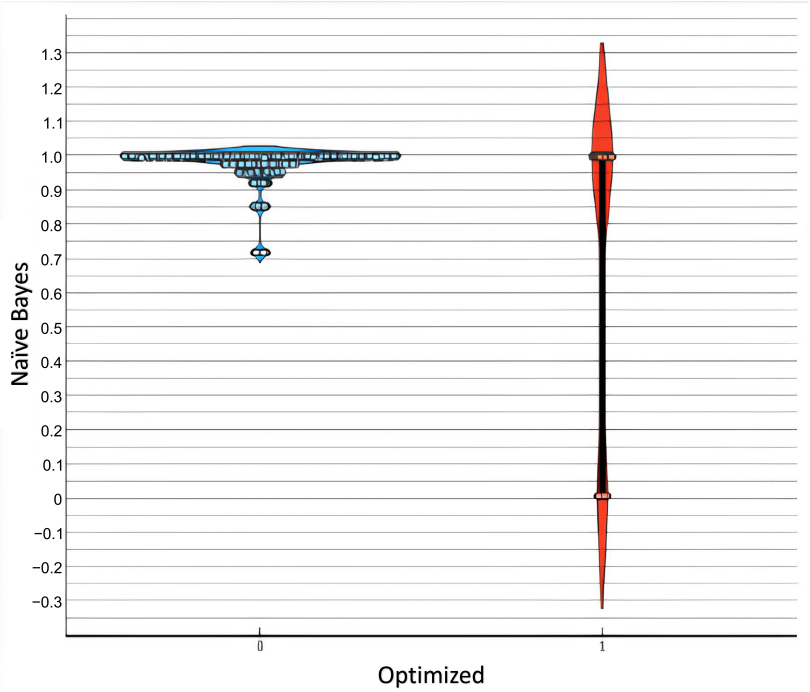


Figure 2. Distribution of performance scores for naive Bayes.

Furthermore, as shown in Figure 4, the distribution of logistic regression performance scores highlights a wider spread of performance thresholds compared to the other models tested. Unlike neural network-based approaches, this distribution shows greater variability, suggesting a class-discrimination ability that strongly depends on the choice of the decision threshold. Compared to the neural network, the curve associated with logistic regression indicates less precise coverage of the best performance thresholds, which can be attributed to a potentially lower area under the curve (AUC). This reflects a slightly lower ability to effectively differentiate between classes. However, this approach remains attractive due to its simplicity and interpretability, which can be assets in certain application contexts.

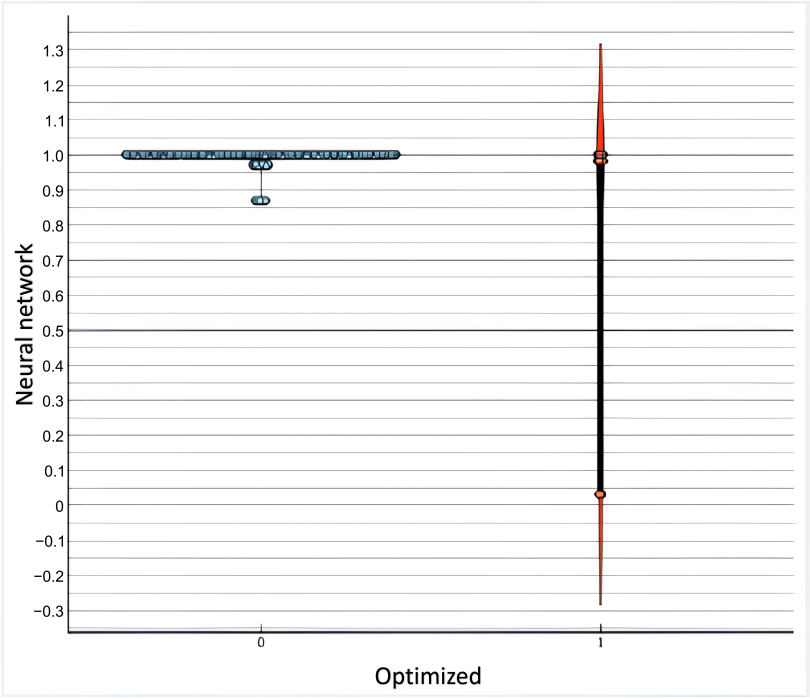


Figure 3. Distribution of performance scores for neural network.

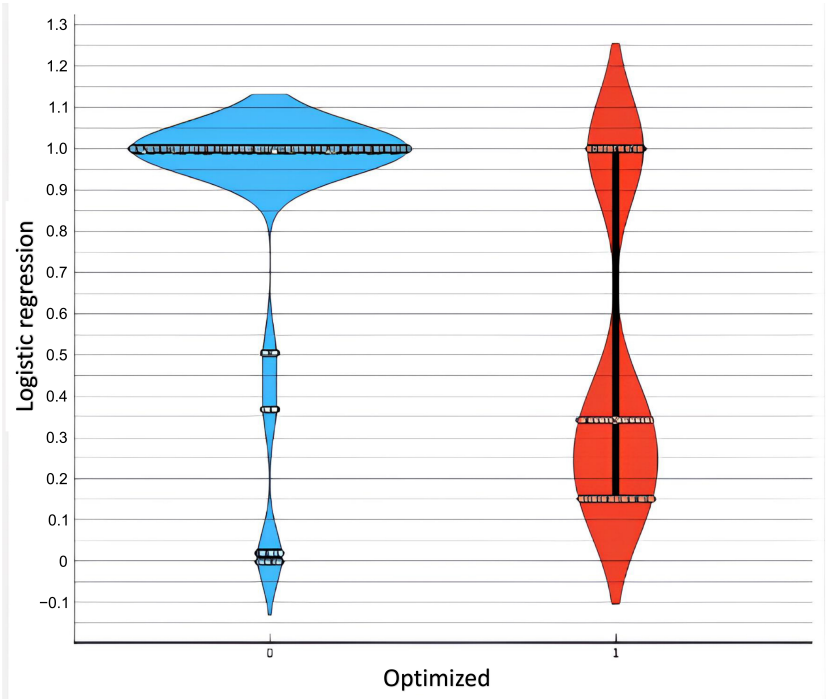


Figure 4. Distribution of performance scores for logistic regression.

Moreover, analysis of the distribution of performance scores for SVM, as shown in Figure 5, reveals a relatively well-structured distribution of optimal thresholds. Unlike more flexible models, such as neural networks, the SVM tends to generate a more rigid classification, resulting in a concentration of performance around certain specific thresholds. Compared to logistic regression, the SVM appears to better capture the separation between classes, suggesting increased robustness in certain scenarios. However, this rigidity may also limit its ability to adapt to complex data variations. Depending on the metric evaluated, it could display a trade-off between accuracy and generalization, positioning it as an

intermediate choice between simple linear models and more complex architectures, such as neural networks.

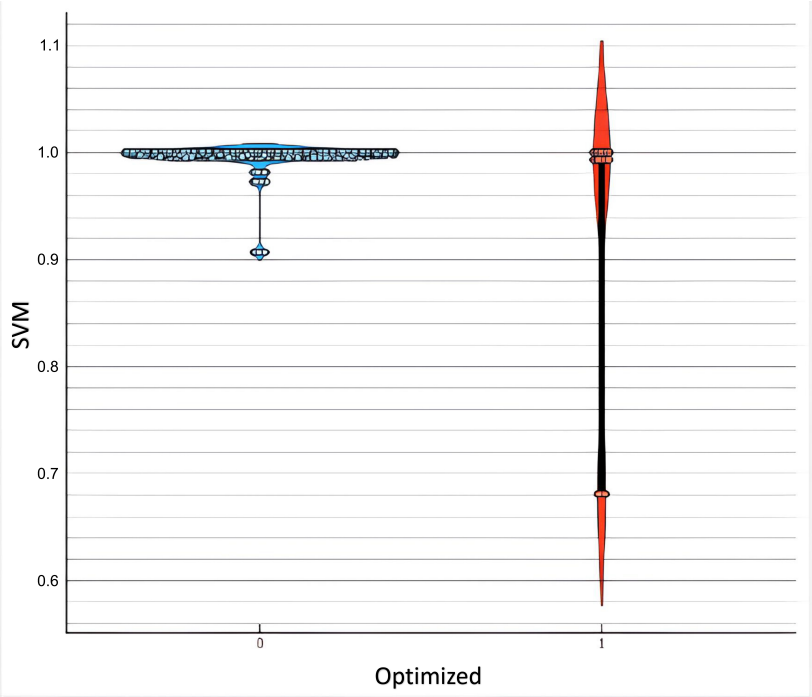


Figure 5. Distribution of performance scores for SVM.

Finally, examining the distribution of performance scores for gradient boosting reveals a balanced spread of optimal values. As shown in Figure 6, this model, known for its ability to reduce errors by combining multiple decision trees, demonstrates a distribution of scores that seems to indicate good performance stability.

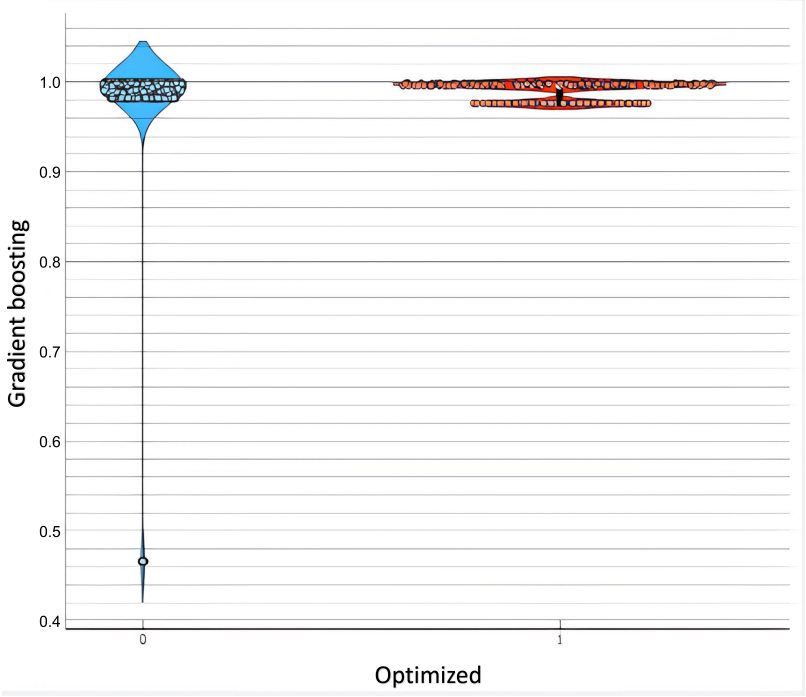


Figure 6. Distribution of performance scores for gradient boosting.

As a result, the neural network's enhanced classification capability is evident in its better detection of both positive and negative classes. This allows the neural network to

discriminate more effectively between classes, thus ensuring higher accuracy and reliability in its predictions than the naive Bayes model.

4.2. Discussion

The results obtained clearly demonstrate the superiority of certain models in transport route optimization. Indeed, approaches based on neural networks and Bayesian methods stand out for their ability to effectively discriminate between different classes, thus ensuring reliable identification of optimal routes. These models, thanks to their ability to integrate a variety of criteria, illustrate the value of using advanced techniques to improve the quality of decisions in transport contexts. In addition, linear models, although offering good class separation capacity, exhibit greater variability related to the choice of the decision threshold. This characteristic highlights their sensitivity to calibration parameters, which can influence the consistency of recommendations in varied operational environments. The simplicity of these approaches nevertheless remains an asset in contexts requiring increased interpretability of results. Finally, certain methods, such as support vector machines and ensemble techniques demonstrate an attractive compromise between robustness and adaptability. Their intermediate performance demonstrates their potential to provide balanced solutions, combining accuracy and the ability to manage the complexity of transportation data. Thus, all the models studied provide a complementary perspective on how to approach route optimization, with each method highlighting specific aspects of the problem.

4.3. Limitations

Despite the promising performance achieved, the current approach presents several limitations intrinsic to the methods used. One of the main challenges lies in the dependence on the quality and representativeness of the training data, which condition the generalization of the models to real-life situations. Furthermore, the variability of operational conditions, such as infrastructure and environmental hazards, can compromise the robustness of the predictions provided. Moreover, some techniques, while effective, require significant computational resources and fine parameter calibration, which can limit their deployment in environments requiring real-time responsiveness. These methodological constraints highlight the need for a rigorous evaluation of the trade-offs between complexity, computational cost, and accuracy of results.

4.4. Comparison with Other Approaches

- "X" indicates the use of a standard number of features and a typical dataset size used in the related works.
- "XXX" in our work refers to a significant enhancement—we expanded the number of features and the dataset size by a factor of 3, aiming to improve model generalization and accuracy.

Comparative analysis of the studies in Table 3 shows the growing importance of machine learning in transport route optimization. Each article highlights specific aspects of the application of these technologies, ranging from route optimization and traffic management to real-time prediction of traffic conditions. The different algorithms used, such as neural networks, genetic algorithms, and traffic simulation models, show great diversity in the methodological approaches adopted.

Ref. [1] explores the application of neural networks and genetic algorithms to improve the sustainability and efficiency of transportation systems. The results indicate that integrating these technologies enables routes to be optimized by considering multiple criteria simultaneously, which is essential for meeting the complex challenges of modern

transportation. Ref. [2] demonstrates the effectiveness of real-time optimization algorithms, underlining the importance of developing robust solutions that adapt to complex urban environments.

Table 3. Comparison of our approach with other works.

Article	Machine Learning	Features	Dataset	Best Algorithm
[1]	X	X	X	Neural networks
[2,3,5]	X	X	X	Not specified
[6,7,11]	X	X	X	Not specified
[10]	X	X	X	Neural networks
Our work	X	XXX	XXX	Neural networks

The work by [3] provides a detailed analysis of urban mobility trends and challenges but does not specify the exact algorithms used, focusing instead on recommendations for reducing urban congestion. Similarly, the ref. [5] study of urban route optimization based on cost, time, and safety highlights the use of path-finding algorithms, with promising results for improving safety and contingency management.

The work by [6,7] shows how supervised and machine learning algorithms can be used for dynamic route optimization strategies and predictive traffic management. These studies highlight the importance of real-time data for maximizing the efficiency of transportation systems. Refs. [10,11] highlight the ability of neural networks and other machine learning algorithms to predict traffic conditions in real-time, improving route planning and optimization.

Our work is distinguished by diverse machine learning algorithms, including support vector machines (SVMs), naive Bayes, neural networks, logistic regression, and gradient boosting. This multicriteria approach enables finer, more precise route optimization by integrating Big Data captured in real-time and exploited by deep learning algorithms to refine optimization strategies further. The results show that neural networks stand out as the best-performing algorithm, with significant potential for improvement thanks to real-time data integration.

5. Conclusions and Future Work

This study proposes an innovative approach for transport route optimization, leveraging machine learning techniques to assess key factors, such as safety, cost, and logistics. Our model, specifically designed for the Dakhla (Morocco) to Paris (France) route, leverages a rich dataset and an advanced algorithm to improve decision-making. The results show significant gains in efficiency and safety compared to traditional methods.

Our analysis highlights the substantial benefits of integrating neural networks into route optimization. With an accuracy of 93.3%, this model demonstrates a strong ability to process and analyze large volumes of data, leading to more precise and reliable decision-making. Beyond enhancing route selection, it dynamically adjusts to real-time logistical constraints, providing a more adaptive and efficient alternative to traditional methods. The continued development of our model seeks to address current transportation challenges while promoting safer and more sustainable mobility solutions. By progressively incorporating advanced technologies and refining our algorithms, we aim to deliver a robust and high-performing system capable of meeting the increasing demands of modern transportation networks.

Future research will focus on integrating real-time data from sensors and connected devices. Leveraging these data with deep learning algorithms would refine optimization

strategies and make them more responsive to dynamic conditions. This development would significantly improve the accuracy and resilience of the transportation network.

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Abbreviations

The following abbreviations are used in this manuscript:

ANNs	Artificial neural networks
AI	Artificial intelligence
CC BY	Creative Commons Attribution
ML	Machine learning
RF	Random forest
SVM	Support vector machine
SVR	Support vector regression
DL	Deep learning
NV	Naive Bayes
CA	Classification Accuracy Technology

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